

Propositional Logic Homework: Each 10 points

(a) Prove use words:

$$(a \vee b) \wedge (\neg a \vee c) \rightarrow \overline{(b \vee c)}$$

(b)

2. *Propositional Resolution*. Use propositional resolution to show that the following sets of clauses are unsatisfiable.

(a) $\{p, q\}, \{\neg p, r\}, \{\neg p, \neg r\}, \{p, \neg q\}$ ← Proof is given in following slide

(b) $\{p, q, \neg r, s\}, \{\neg p, r, s\}, \{\neg q, \neg r\}, \{p, \neg s\}, \{\neg p, \neg r\}, \{r\}$ ← You prove this

Note $\{p, q\} = p \vee q$ is a clause. We use “ $\sim$ ” for negation operator, “ $\&$ ” for AND operator. Thus (a) means show $(p \vee q) \& (\sim p \vee r) \& (\sim p \vee \sim r) \& (p \vee \sim q)$ is unsatisfiable.

Use resolution to prove that

$(p \vee q) \& (\sim p \vee r) \& (\sim p \vee \sim r) \& (p \vee \sim q)$ is unsatisfiable.

Proof. For the compound proposition to be satisfiable, each of the four clauses must be **valid** for a proper assignment of p,q,r. Thus by resolution, we have

By distribution, $A \& B \& C \& D = D \& A \& B \& C$

Thus the above compound proposition becomes

$$\begin{aligned} & (p \vee \sim q) \& (p \vee q) \& (\sim p \vee r) \& (\sim p \vee \sim r) \\ = & [(p \vee \sim q) \& (p \vee q)] \& [(\sim p \vee r) \& (\sim p \vee \sim r)] \\ = & [p \vee p] \& [\sim p \vee \sim p] \quad \text{do resolution on both} \\ = & [p] \& [\sim p] \quad \text{why?} \\ = & \text{false} \quad \text{why?} \end{aligned}$$

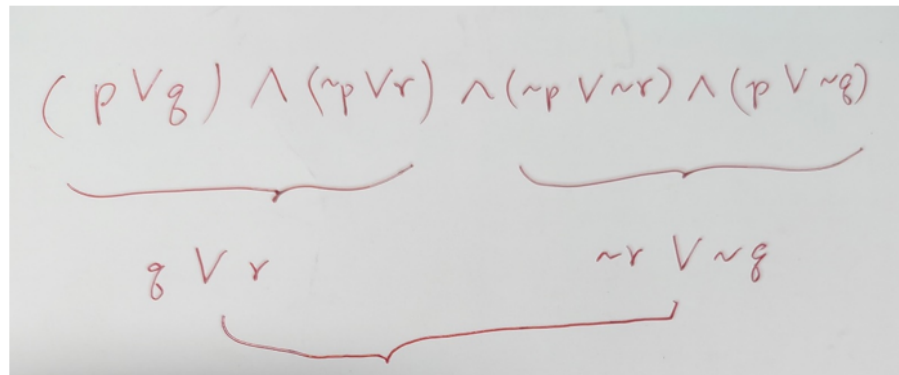
Thus, there is no assignment that makes the proposition true. It is unsatisfiable.

This is another attempt to prove. It did not provide useful things.

Use resolution to prove that

$(p \vee q) \& (\sim p \vee r) \& (\sim p \vee \sim r) \& (p \vee \sim q)$ is unsatisfiable.

Proof. For the compound proposition to be satisfiable, each of the four clauses must be **valid** for a proper assignment of p,q,r. Thus by resolution, we have



The image shows a handwritten resolution process on a piece of paper. At the top, the expression $(p \vee q) \wedge (\sim p \vee r) \wedge (\sim p \vee \sim r) \wedge (p \vee \sim q)$ is written. Below it, two horizontal curly braces are drawn under the first two terms and the last two terms. Under the first brace, the expression $q \vee r$ is written. Under the second brace, the expression $\sim r \vee \sim q$ is written. A third, larger horizontal curly brace is drawn below these two, spanning the width of both $q \vee r$ and $\sim r \vee \sim q$.

$$r \vee \sim r = \text{true}$$

We start with the whole compound resolution is true. We end up with it is true. We did not prove/get anything new.

Propositional Logic Homework: (C)

58. How many of the disjunctions $p \vee \neg q$, $\neg p \vee q$, $q \vee r$, $q \vee \neg r$, and $\neg q \vee \neg r$ can be made simultaneously true by an assignment of truth values to p , q , and r ?

Propositional Logic Homework (D)

In standard mathematics, can the following three equations hold (being true) simultaneously?

$$x = y$$

$$y = z$$

$$z = -x$$

Why?

In propositional logic, consider the following three compound propositions

$$A \Rightarrow B$$

$$B \Rightarrow C$$

$$C \Rightarrow \neg A$$

- (a) They can not be true simultaneously in all cases?
- (b) They can be true simultaneously in some cases?
- (c) They can be true simultaneously in all cases?

Provide your answer. Explain or prove it.

(Hint: use the truth table to guide your thinking.)